

Learning objectives

1. After describing the **morphology of thrombi (including postmortem clots)**, discuss and compare the primary thrombotic diseases which include **Factor V Leiden, Prothrombotic Gene mutation, Homocysteinemia, Anti-thrombin III deficiency, and Protein S and Protein C deficiency**
2. Describe and discuss secondary thrombotic disorders which include **malignancies, etc.** and heparin induced thrombocytopenia, **and anti-phospholipid syndrome**
3. Describe, discuss, compare the **fate of thrombi, venous v. arterial and cardiac thrombi, emboli including pulmonary emboli, systemic emboli, and paradoxical emboli**
4. Describe, discuss and compare other various emboli including **fat emboli, air emboli, and amniotic fluid emboli**

3 Major risk factors for Thrombosis

Virchow's Triad

1. Endothelial Injury

- Atherosclerosis / Hypercholesterolemia / Inflammation
- HTN / Vasculitis / Diabetes
- Toxins (Cigarette smoke) / Elevated Homocysteine

2. Abnormal Blood Flow

- Stasis / Turbulence

3. Hypercoagulable State

- Primary (genetic) or Secondary (acquired) disorders

Endothelial Injury

- **Severe endothelial injury exposes vWF and tissue factor: may trigger thrombosis**
- **Inflammation** promotes thrombosis: shifts gene expression in endothelium to “prothrombotic.” (i.e., **endothelial activation or dysfunction**) – causes include:
 - physical injury
 - infectious agents
 - abnormal blood flow
 - inflammatory mediators
 - metabolic abnormalities (e.g., hypercholesterolemia or homocystinemia)
 - toxins absorbed from cigarette smoke
- **Procoagulant changes:** e.g.,: endothelial cells activated by inflammatory cytokines, downregulate expression of thrombomodulin **procoagulant and pro-inflammatory actions of thrombin enhanced**
- **Antifibrinolytic effects:** activated endothelial cells secrete plasminogen activator inhibitors (PAIs) (limit fibrinolysis and downregulate expression of t-PA) – favors development of thrombi

Turbulence (Abnormal blood flow)

- **Turbulence contributes to arterial and cardiac thrombosis by causing endothelial injury/ dysfunction and forming countercurrents (contribute to stasis)**
- **Turbulence and stasis:**
 - Promote endothelial activation, enhancing procoagulant activity
 - Disrupt laminar flow- bring platelets into contact with endothelium
 - Prevent washout and dilution of activated clotting factors by fresh flowing blood and inflow of clotting factor inhibitors
- **Examples:**
 - **Ulcerated atherosclerotic plaques** expose subendothelial vWF and tissue factor, (also cause turbulence)
 - **Aneurysms** result in local stasis
 - **Acute myocardial infarction** - areas of noncontractile myocardium associated with stasis and flow abnormalities (promote formation of cardiac mural thrombi)

Hypercoagulability

- Abnormal tendency of blood to clot, usually caused by changes in coagulation factors
- Important in venous thrombosis
- Divided into primary (genetic) and secondary (acquired) disorders
- Primary include:
 - **Factor V Leiden**
 - **Prothrombin gene mutation**
 - **Elevated levels of homocysteine**
 - Rare inherited causes of primary hypercoagulability: **deficiencies of anticoagulants (e.g., antithrombin III, protein C, or protein S)**; present with venous thrombosis and recurrent thromboembolism beginning in adolescence or early adulthood

Pathology of Thrombosis

- **Venous thrombi**
 - can cause painful congestion and edema distal to obstruction
 - **may embolize to lungs; almost always occlusive**
 - Form long luminal cast – form in sluggish venous circulation - **usually occur at sites of stasis**
 - aka **red thrombi**; firm, focally attached to vessel wall, contain lines of Zahn
 - **veins of lower extremities most commonly involved (90% of cases)**
- **Arterial thrombi**
 - can also embolize; cause downstream infarctions; often occlusive
 - most common sites: **coronary, cerebral**, and femoral arteries (decreasing order of frequency)
 - **Arterial or cardiac thrombi usually begin at sites of turbulence or endothelial injury**
 - Thrombi in heart chambers **aka mural thrombi, esp. seen in:**
 - abnormal myocardial contraction (e.g., arrhythmias, dilated cardiomyopathy, myocardial infarction)
 - endomyocardial injury (myocarditis or catheter trauma)
- **Postmortem clots** - gelatinous with dark-red dependent portion (red cells settled by gravity) and yellow “chicken fat” upper

Venous Thrombosis (Phlebothrombosis)

- **Most occur in superficial or deep veins of leg**
- **Superficial venous thrombi usually occur in saphenous veins in setting of varicosities; can cause local congestion, swelling, pain, and tenderness, rarely embolize**
- **Deep venous thrombus involving large leg veins (e.g., popliteal, femoral, and iliac)— more often embolize to lungs; may cause pulmonary infarction**
- DVTs may cause local pain and edema due to venous obstruction, but because of venous collateral channels, **about 50% of DVTs asymptomatic; recognized only after embolization**
- Lower extremity DVTs often associated with hypercoagulable states; common predisposing factors - **bed rest and immobilization, congestive heart failure; trauma, surgery, and burns** cause immobilization and also vascular insults

Primary (Genetic)

Robbins and Cotran, Pathological Basis of Disease, 10th ed., 2020, Ch.4, Table 4.2 Hypercoaguable states

Common

- **Factor V mutation: factor V Leiden** (Arg to Gln substitution in amino acid residue 506 leading to resistance to activated protein C)
- **Prothrombin mutation** (G20210A noncoding sequence variant leading to increased prothrombin levels)
- Increased levels of factors VIII, IX, XI, or fibrinogen (genetics unknown)

Rare

- **Antithrombin III deficiency**
- **Protein C deficiency**
- **Protein S deficiency**

Very Rare

- Fibrinolysis defects
- Homozygous homocystinuria (deficiency of cystathione β -synthetase)

Secondary (Acquired)

Strong Risk Factors for Thrombosis

- **Prolonged bed rest or immobilization**
- Myocardial infarction
- Atrial fibrillation
- Tissue injury (surgery, fracture, burn)
- **Cancer**
- Prosthetic cardiac valves
- Disseminated intravascular coagulation
- **Heparin-induced thrombocytopenia**
- **Antiphospholipid antibody syndrome**

Other Risk Factors for Thrombosis

- Cardiomyopathy
- Nephrotic syndrome
- **Hyperestrogenic states (pregnancy and postpartum)**
- **Oral contraceptive use**
- Sickle cell anemia
- **Smoking**

Pathology of Thrombosis

Classic presentation – recurrent DVTs or DVT at young age

Primary (Genetic)

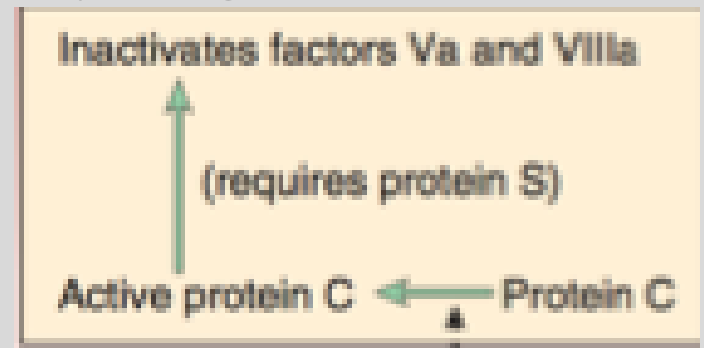
- Factor V Leiden
- Prothrombin Gene Mutation
- Homocysteinemia
- Anti-Thrombin III Deficiency
- Protein C Deficiency
- Protein S Deficiency

Secondary (Acquired)

- Heparin Induced Thrombocytopenia (HIT)
- Anti-Phospholipid Syndrome (APLS)
- Others: Prolonged bed rest, MI, Afib, Cancer, DIC, Post-Op, Prosthetic cardiac valves

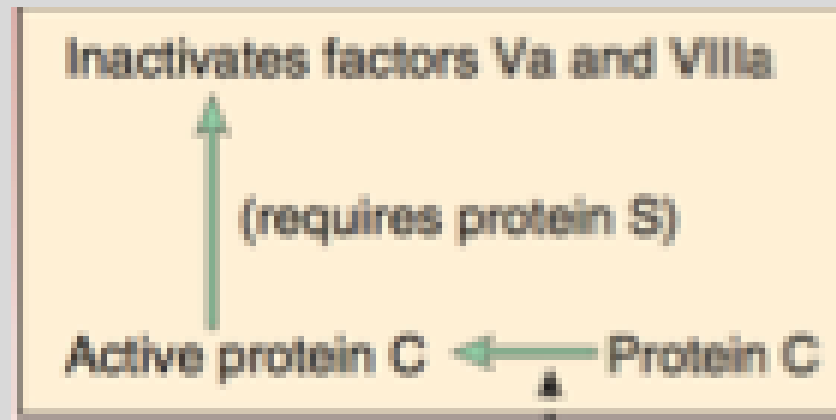
FACTOR V LEIDEN

- **Most common hereditary thrombosis syndrome**, 2% to 15% of Caucasians are carriers
- **Autosomal Dominant** with incomplete penetrance
- Single nucleotide mutation makes **Factor V resistant to cleavage** and inactivation **by protein C** (loss of antithrombotic regulator)
- Heterozygotes: 5X increased risk venous thrombosis, homozygotes 50X increase
- Factor V Leiden heterozygote (usually asymptomatic) may develop DVTs when combined with inactivity (e.g., prolonged airplane travel)
- Mutation frequency near 60% in patients with recurrent DVTs
- Patients < 50 years old with thrombotic events should be considered for any clotting disorder, especially Factor V Leiden — even when acquired risk factors are present
- Diagnosis: Activated Protein C (APC) Assay (failure to prolong PTT with addition of purified protein C) or **genetic testing**



Protein C & S Deficiency

- Natural anticoagulants; made by liver, need to be activated
 - Vitamin K-dependent proteins that serves as a cofactor for activated protein C to inactivate the procoagulant factors Va and VIIIa, thereby reducing thrombin generation
 - **Mutation results in inability to inactivate factors V & VIII; results in increased thrombin formation**
 - Individuals have increased risk of thrombotic skin necrosis with warfarin
- CLINICAL CORRELATE: if patient is placed on warfarin therapy, the liver will not produce factors 2,7,9,10, protein C and S. All pre-existing protein C and S will degrade first leaving pre-existing factors 2,7,9,10 to be relatively high
- Need Heparin bridge
 - Presents with venous thrombosis and recurrent thromboembolism beginning in adolescence/early adulthood



Secondary (Acquired) Hypercoagulable (Clotting) Disorders

- Multifactorial
- Prolonged bed rest
- MI
- Atrial fibrillation
- DIC
- Post-op
- Prosthetic cardiac valves
- Malignancy (disseminated) – release of prothrombotic mediators
- Smoking or obesity
- Oral Contraceptives / Pregnancy – Estrogen increases coagulation factors and decreases ATIII
- Advanced Age – Decreased endothelial PGI₂ production
- Hyperviscosity – Polycythemia / Waldenstroms macroglobulinemia
- **Heparin Induced Thrombocytopenia (HIT)**
- **Anti-Phospholipid Antibody Syndrome (APLS)**

Heparin-Induced Thrombocytopenia (HIT) Syndrome

- **Thrombocytopenia** most common manifestation (about 5% of persons receiving unfractionated heparin (see Path of Hemorrhage lecture); **thrombosis** most serious complication occurring in ~50% of cases; affects both veins and arteries
- Diagnosis: demonstration of anti-PF4-heparin antibodies
- Low-molecular-weight heparin preparations induce HIT less frequently
- May present with sudden onset of pain, redness, and swelling of an arm or leg; ecchymosis, necrosis of skin, gangrene, stroke or MI may develop
- 2 Types:
- HIT Type I: (non-immune) asymptomatic transient drop of platelets, returns to normal
- HIT Type II: (immune mediated) Clinically serious condition
- Thrombosis may occur in either veins or arteries

Antiphospholipid Antibody Syndrome (APS)

- Autoimmune disorder characterized by:
 - Presence of one or more antiphospholipid (aPL) autoantibodies
 - Venous or arterial thromboses, or pregnancy complications (e.g., recurrent miscarriages, unexplained fetal death, and premature birth.)
- Primary or secondary (e.g., **lupus**)
- Primary APS: patients only have hypercoagulable state and no evidence of autoimmune diseases
- Presence of Antiphospholipid autoantibodies
 - Anticardiolipin antibodies (IgG or IgM)
 - Anti-beta-2-glycoprotein-I antibodies (IgG or IgM)
 - Lupus anticoagulants
- Various clinical manifestations: **recurrent thromboses, repeated miscarriages, cardiac valve vegetations, and thrombocytopenia**
- Clinical presentations can include **pulmonary embolism (following lower extremity venous thrombosis), pulmonary hypertension (from recurrent subclinical pulmonary emboli), valvular heart disease, stroke, bowel infarction, or renovascular hypertension**

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch.4

Antiphospholipid Syndrome - Pathogenesis

- Not fully understood: **APL antibodies directed against anionic membrane phospholipids or proteins associated with phospholipids** (include cardiolipin and β 2 -glycoprotein I)
- **Anti- β 2 -glycoprotein antibodies suspected to have major role in APS by activating endothelial cells, monocytes, and platelets**
- Patients with APS also show evidence of complement activation and inhibition of fibrinolytic processes: both favor **prothrombotic state**
- Pregnancy morbidity: **defining manifestation of APS; fetal loss not thought to be caused by thrombosis, but from antibody-mediated interference with growth and differentiation of trophoblasts: leads to failure of placentation**

APL Antibodies

- Antiphospholipid antibodies also identified in 5% to 15% of normal individuals, implying Abs not sufficient to cause full-blown syndrome
- Antibodies induce hypercoagulable state in vivo- but interfere with phospholipids and inhibit coagulation in vitro, **prolonging the PTT**
- Mixing study: patient plasma mixed with control plasma does not correct aPTT
- Often **false-positive serologic test for syphilis** because the antigen in standard assay is embedded in cardiolipin, which cross reacts with phospholipids of *Treponema pallidum*
- Diagnosis based on clinical features and demonstration of APL antibodies in serum
- Therapy - various forms of anticoagulation

Robbins and Cotran Pathologic Basis of Disease, 10th ed., 2021

Other Secondary Causes

- **Prolonged Bed Rest:** reduced venous return from the legs leads to stasis
- **Oral Contraceptives** or Hyperestrogenic state from **pregnancy** increase hepatic synthesis of coagulation factors and reduce anticoagulant synthesis
- **Myocardial Infarction:** area of infarct has poor contracting heart muscle leading to stasis
- **Atrial Fibrillation:** stasis of blood in the left atrial appendage
- **Malignancy:** tumors release procoagulants into the blood stream
- **Hyperviscosity** (such as polycythemia vera): high concentrations of prothrombotic microparticles, stasis, and activation of platelets, leukocytes, and endothelial cells; stasis
- **Old age:** reduced endothelial PGI 2 production
- **Post-Op:** vascular damage and prolonged bed rest
- **Prosthetic Cardiac Valves**

FATE OF THROMBUS

- **Propagation:** additional platelets/fibrin attach and increase the size of thrombus
- **Embolization:** thrombi dislodge, travel to other sites in vasculature
- **Dissolution/Resolution:** result of fibrinolysis, which can lead to rapid shrinkage and total disappearance of recent thrombi
 - extensive fibrin deposition and cross-linking in **older thrombi: more resistant to lysis** (therapeutic administration of fibrinolytic agents such as t-PA (e.g., in acute coronary thrombosis) usually effective only when given during first few hours of thrombotic event)
 - **Organization and recanalization:** older thrombi become organized by ingrowth of endothelial cells, smooth muscle cells, and fibroblasts; capillary channels eventually form within thrombus which allows blood flow

FAT EMBOLISM SYNDROME

- Symptomatic patients (minority) - **pulmonary insufficiency, neurologic symptoms, anemia, thrombocytopenia**
- **Diffuse petechial rash; fatal in 5% to 15% of cases**
- **Thrombocytopenia** because of platelet adhesion to fat globules and subsequent aggregation or splenic sequestration
- **Anemia** can result from red cell aggregation and/or hemolysis
- Usually, 1 to 3 days after injury - sudden onset of tachypnea, dyspnea, and tachycardia; irritability; restlessness, can progress to delirium or coma

Air Embolism

- Occurs when communication between vasculature and outside air exists; negative pressure gradient “sucks” in air
- **Air may be introduced during endovascular and invasive procedures, mechanical ventilation;** large volume of air (**more than 100 mL**) needed to cause effect in pulmonary circulation
- Introduction of 300 to 500 mL of air at 100 mL/sec may be fatal
- Coronary Bypass Surgery, Laparoscopic Abdominal or Obstetric Surgery, Central Line Placement, Chest wall injuries, Scuba & Deep sea divers, Underwater Construction
- **Air in pulmonary vasculature blocks perfusion of downstream region; and microemboli of air trapped in pulmonary capillaries cause intense inflammatory response with release of cytokines that can injure alveoli**
- Bubbles in central nervous system can cause mental impairment and sudden onset of coma
- **Classic presentation: sudden collapse, neurologic symptoms that mimic stroke, apnea, and even cardiovascular collapse**

Decompression sickness

- Form of gas embolism - patients experience sudden decreases in atmospheric pressure
- Scuba, deep sea divers, and underwater construction workers
- When air is breathed at high pressure (e.g., deep sea dive), increased amounts of gas (especially nitrogen) dissolve in blood and tissues; if diver **ascends (depressurizes)** too rapidly, **nitrogen comes out of solution in tissues and blood**
- **Rapid formation of gas bubbles within skeletal muscles and supporting tissues in/ around joints: causes pain (the bends)**
- **In lungs, gas bubbles in vasculature cause edema, hemorrhage, and focal atelectasis or emphysema; respiratory distress (the chokes)**
- Chronic form of decompression sickness: **Caisson disease: persistence of gas emboli in skeletal system leading to many foci of ischemic necrosis**; more common sites: femoral heads, tibia, and humeri
- Acute decompression sickness: treated by putting patient in a chamber under enough high pressure to force gas bubbles back into solution; subsequent slow decompression allows gradual resorption and exhalation of gases, preventing obstructive bubbles from reforming

Summary Slide

- Morphology of thrombi – lines of Zahn v. post-mortem clots, Virchow's triad (endothelial injury, abnormal blood flow, hypercoagulable state)
- Primary thrombotic (clotting) disorders
 - Factor V Leiden – AD, most common, Factor V resistant to cleavage by protein C
 - Prothrombin gene mutation – AD, increase prothrombin, increased thrombin
 - Homocysteinemia – endothelial damage, prothrombotic effects
 - Anti-thrombin III deficiency – def. leads to increased thrombin
 - Protein C and S deficiencies – inability to inactivate factors V and VIII: results in increased thrombin
- Secondary thrombotic (clotting) disorders
 - Malignancies, OCP, pregnancy, age, hyperviscosity
 - Heparin induced thrombocytopenia – induced IgG autoAbs to heparin complexes and platelet factor 4
 - Anti-Phospholipid Ab syndrome – recurrent thrombosis, repeated miscarriages, decreased platelets
- Fate of thrombi – propagation, recanalization, embolization, dissolution
- Venous v. arterial and cardiac thrombi
- Emboli – pulmonary emboli, paradoxical emboli, systemic emboli, fat emboli, air emboli, and amniotic fluid emboli

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